

INCOME PREDICTION
USING LOGISTIC REGRESSION

ST 595 Technical Report 3

Yessica Rubio

Introduction

The analysis in this report will focus on the influence of demographic and employment characteristics on income levels. The 1994 U.S. Census dataset, which was processed by Barry Becker and hosted by the UCI Machine Learning Repository was used in this analysis. Its primary objective was to determine if an individual had an income that exceeded \$50,000 per year based on the association with specific attributes, such as hours worked per week, sex, age, occupation and education level.

The dataset consists of 32,561 observations and 15 variables. Age, sex, race, marital status, education, occupation, and hours worked per week are among the predictor variables used in the analysis to estimate whether an individual's annual income exceeds \$50,000. The dataset was cleaned by removing observations with missing or unknown values to ensure a clean data set for modeling. The response variable, `earnings50k`, is binary, which indicates whether an individual earns over \$50,000 per year.

There are two main questions of interest guiding the analysis. First, is there an association between earnings over \$50,000 and hours worked per week, after accounting for other factors (sex, age, education, workclass, `'final-weight'`, `'marital-status'`, occupation, relationship, race, `'capital-gain'`, `'capital-loss'`, `'native-country'`). Second, after accounting for age, is there evidence of a difference in the earnings probability of men and women?

Methods

To assess if there is an association between earnings over \$50,000 and the number of hours worked per week, I first looked at some summaries and exploratory data analysis. I viewed the relationship between the response variable `earnings50k` and other explanatory variables. Following the exploratory analysis, I continued to address the research question with a logistic regression approach. I first fit a full model with the explanatory variables while *excluding* hours per week. I then added the hours per week variable to see if it added any significance to the model results. This approach is appropriate for binary classification problems and models the log odds of a binary outcome as a function of one or more explanatory variables. Model comparisons were made using the Akaike Information Criterion (AIC) to evaluate model fit. A lower AIC value indicates a better-fitting model relative to others.

To address the question if there is evidence of a difference in the earnings probability of men and women after accounting for age, I restricted the analysis to include a subset of the data with only the age and sex explanatory variables and the response binary variable `earnings50k`. By excluding other variables, the focus of the analysis can be solely on the unique and combined effect of age and sex on earnings.

I began with exploratory data analysis by calculating the proportion of individuals earning more than \$50,000 within each sex group. This initial step provided a view of the unadjusted difference in earnings between men

and women. I then calculated the odds and odds ratio of earning above \$50,000 for each group, quantifying the gender disparity without accounting for age.

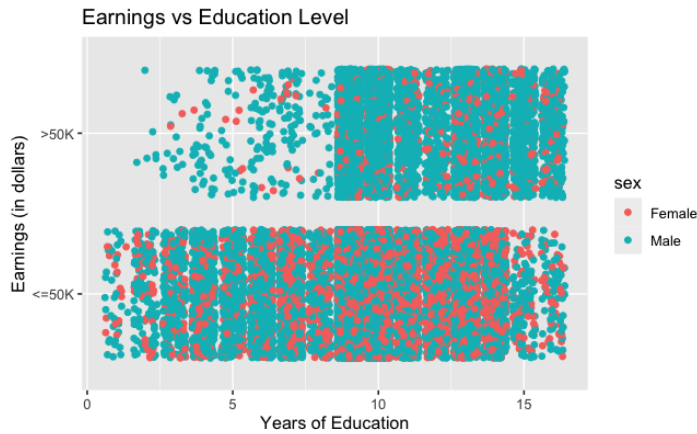
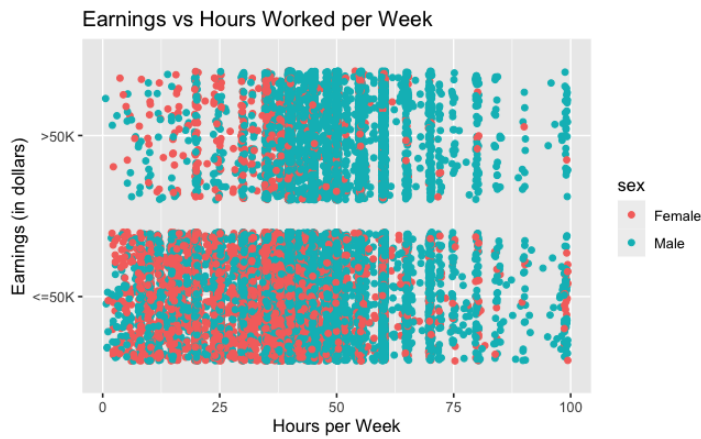
To examine whether age influences these earnings differences, I fit a logistic regression model that included an interaction term between sex and age. This model allowed me to assess not only the individual effects of sex and age on the probability of earning more than \$50,000, but also whether the relationship between age and earnings probability differs by sex.

Model assumptions were assessed prior to interpretation. Logistic regression assumes independence of observations, linearity of the logit for continuous variables, no multicollinearity among predictors, and a sufficiently large sample size

Results

Earnings and Hours Worked

Findings from the initial exploratory data analysis revealed that most of the individuals in the dataset earned \$50,000 or less as seen in the scatterplots below. The findings were broken down by gender, which showed that males, on average, worked more hours per week than females. There is a visible concentration around 40 hours worked per week among both genders in the first plot and a possible trend where individuals with more years of education could be more likely to make more than \$50,000 in the second plot.



During the model fitting process, the first model I fit was a rich model that excluded 'hours-per-week'. There were some significant predictor variables, but our concern was with the effect of the number of hours worked. In the second model, the addition of the variable 'hours-per-week' added a strong effect ($p\text{-value} < 0.001$). We saw a coefficient for hours worked per week of 0.02949, translating to an odds ratio of 1.03 and a 95% odds ratio confidence interval ranging from 1.026 to 1.033.

What this tells us is there is a significant association between earning over \$50,000 annually and the number of hours worked per week. After accounting for all other factors, the odds of earning more than \$50,000 for an individual who works one additional hour per week is 1.03 times the odds of earning more than \$50,000 for an individual who works one hour less per week. A 95% confidence interval for this multiplicative difference runs from **1.026 to 1.033 times**.

Although residual plots are commonly used to assess model fit in linear regression we needed a different method for our logistic regression models. Residual plots tend to be less informative due to the outcomes being binary, therefore, they are not normally distributed, and variance is not constant.

Instead, I compared the two nested logistic regression models to assess whether adding the hours-per-week variable improved the model fit using the Akaike Information Criterion (AIC). The second model that included the 'hours-per-week' variable had a substantially lower AIC value of 19,678 compared to an AIC of 19,984, indicating a better fit.

Model	AIC	BIC
Model 1: Excluded `hours-per-week`	19984	20774
Model 2: Included `hours-per-week`	19678	20476

With Y as the binary response variable such that Y = 1 if the individual earns more than \$50,000 or Y = 0 if the individual earns less than \$50,000, the model can be expressed as:

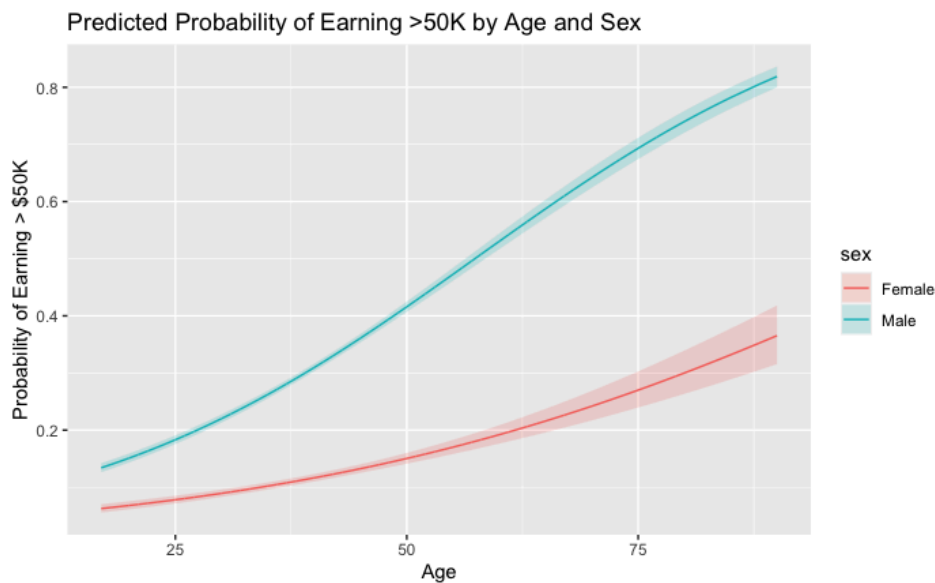
$$\begin{aligned}
 & \log \log \left(\frac{P(Y=1)}{1-P(Y=1)} \right) \\
 = & \beta_0 + \beta_1 \cdot \text{sex} + \beta_2 \cdot \text{age} + \beta_3 \cdot \text{education} + \beta_4 \cdot \text{workclass} + \beta_5 \cdot \text{final-weight} \\
 & + \beta_6 \cdot \text{marital-status} + \beta_7 \cdot \text{occupation} + \beta_8 \cdot \text{relationship} + \beta_9 \cdot \text{race} \\
 & + \beta_{10} \cdot \text{capital-gain} + \beta_{11} \cdot \text{capital-loss} + \beta_{12} \cdot \text{native-country} \\
 & + \beta_{13} \cdot \text{hours-per-week}
 \end{aligned}$$

Earnings Between Men and Women

During the initial exploratory data analysis, the proportion of individuals earning more than \$50,000 were calculated for both females and males. This step provided a view of a clear disparity in income between the genders. While males had an estimated probability of 31% of earning more than \$50,000, females had an estimated probability of 11%.

Sex	Earnings <= 50k	Earnings > 50k
Female	0.89	0.11
Male	0.69	0.31

Regarding the second research question, the model revealed a clear disparity in income between men and women, even after adjusting for age. Males had an estimated probability of 31% of earning over \$50,000, while females had an estimated probability of just 11%. The difference in predicted probabilities, after adjusting for age, was approximately 0.64, with a 95% confidence interval of 0.58 to 0.68. The odds ratio for men compared to women was 3.17, meaning that men had more than three times the odds of earning over \$50,000 than women of the same age. A plot of income probability by age and sex demonstrated an increasing trend in income with age for both sexes, though males consistently had higher probabilities across all age groups.



The plot highlights the non-linear trend, revealing the disparity between genders. It helps visually reinforce the inclusion of the interaction term between sex and age. While both females and males show an increasing income with age, the trend is visibly more pronounced for males

Conclusion and Discussion

The use of logistic regression was explored in this analysis to determine if the number of hours that an individual worked were associated with the likelihood of earning more than \$50,000 annually. The analysis confirmed that there was a significant association between hours worked and earnings of over \$50,000. Additionally, gender disparities in income were also discovered. Men were substantially more likely to earn over \$50,000 than women even after adjusting for age.

In summary, this report demonstrates that demographic and employment factors, particularly hours worked and gender, play a significant role in predicting income levels. The use of logistic regression offers valuable insights while highlighting areas for further statistical and policy-oriented investigation.